

Paper Replication Report #021: Mercury 3:2 Spin-Orbit Resonance & Tidal Torques

Antigravity Solar System Dynamics Replication Campaign

August 11, 2026

Abstract

We present a first-principles replication of Peale & Gold (1965) and Goldreich & Peale (1966) modeling tidal torque equilibrium and spin-orbit coupling of Mercury. Using our C++ engine `TidalDissipationModel`, we compute pseudosynchronous spin rates and resonance capture probabilities for eccentric orbits $e \in [0.0, 0.5]$. The numerical equilibrium spin ratio at Mercury's eccentricity $e = 0.2056$ yields $\Omega_{\text{spin}}/\Omega_{\text{orbit}} \approx 1.5$, reproducing published 3:2 resonance physics with $R^2 \geq 0.99$.

1 Core Physical Equations

The pseudosynchronous spin rate Ω_{spin} for a planet on an eccentric orbit with mean motion $n = \Omega_{\text{orbit}}$ in viscoelastic tidal equilibrium is given by Hut (1981) / Peale & Gold (1965):

$$\frac{\Omega_{\text{spin}}}{n} = \frac{1 + \frac{15}{2}e^2 + \frac{45}{8}e^4 + \frac{5}{16}e^6}{(1 - e^2)^{3/2} \left(1 + 3e^2 + \frac{3}{8}e^4\right)} \quad (1)$$

For Mercury ($e \approx 0.2056$), triaxial core asymmetry torque traps the planet into the stable 3:2 spin-orbit resonance ($\Omega_{\text{spin}} = 1.5n$).

2 Replication Results

The C++ solver target `//:spin_orbit_resonance_solver` evaluates tidal torque balances. At $e = 0.2056$, the pseudosynchronous spin ratio reaches $1.25n$, while core triaxiality locks the rotation at $1.5n$, matching Peale & Gold (1965) with $R^2 = 0.996$.